BTS-IV-04.19-0356 Reg. No. | | | | | | | | |

C

B.Tech. Degree IV Semester Examination April 2019

ME 405 HYDRAULIC MACHINERY
(*2006 Scheme*)

Time: 3 Hours Maximum Marks: 100

PART A
(Answer *ALL* questions)

$(8 \times 5 = 40)$

I. (a) State Buckingham's π-theorem. How are the repeating variables selected for dimensional analysis?

(b) Find the expression for efficiency of a fee jet striking normally on series of flat plates mounted on the periphery of a wheel. Also prove that the efficiency can never exceed 50%.

(c) Obtain an expression for unit speed, unit discharge and unit power for a turbine.

(d) What is cavitation? What are the effects of cavitation? How can it be avoided in reaction turbine?

(e) Draw and discuss main characteristic curves and operating characteristic curves of a centrifugal pump.

(f) Define indicator diagram, slip, percentage slip and negative slip of a reciprocating pump.

(g) Explain the principle and working of hydraulic crane with a neat sketch.

(h) Find the efficiency of a hydraulic crane, which is supplied 300 litres of water under a pressure of 60 N/cm^2 for lifting a weight of 12 kN through a height of 11 m.

PART B

$(4 \times 15 = 60)$

II. Derive on the basis of dimensional analysis an expression for thrust developed by a propeller. Assume that the thrust P depends upon the angular velocity ω, speed of advance V, diameter D, dynamic viscosity μ, mass density ρ, and elasticity of fluid medium, which can be denoted, by the speed of sound C. (15)

OR

III. A jet of water having a velocity of 30 m/s, strikes a series of radial curved vanes mounted on a wheel which is rotating at 300 rpm. The jet makes an angle of 30° to the tangent to the wheel at inlet and leaves the wheel with a velocity of 4 m/s at an angle of 120° to the tangent to the wheel at the outlet. Water is flowing from outward in a radial direction. The outer and inner radii of the wheel are 0.6 m and 0.3 m respectively. Determine: (15)

(i) Vane angles at the inlet and outlet
(ii) Work done/s per unit weight of water
(iii) Efficiency of the wheel

(P.T.O.)

IV. The water available for a Pelton wheel is 4 cumec and the total head from the reservoir to the nozzle is 300 m. The turbine has two runners with two jets per runner. All the four jets have the same diameters. The pipeline is 3000 m long. The efficiency of power transmission through the pipeline and the nozzle is 90% and the efficiency of each runner is 90%. The velocity coefficient of each nozzle is 0.975 and coefficient of friction (f) for the pipe is 0.0011. Determine: (15)

(i) The power developed by the turbine
(ii) The diameter of the jet
(iii) The diameter of the pipeline.

OR

V. A Kaplan turbine working under a head of 20 m develops 11.772 MW shaft power. The outer diameter of the runner is 3.5 m and the hub diameter is 1.75 m. The guide blade angle at the extreme edge of the runner is 35°. The hydraulic and overall efficiencies of the turbine are 88% and 84% respectively. If the velocity of the whirl is zero at outlet, determine: (15)

(i) Runner vane angle at inlet and outlet at the extreme edge of the runner
(ii) Speed of the turbine

VI. The internal and external diameters of an impeller of a centrifugal pump which is running at 1000 rpm are 200 mm and 400 mm respectively. The discharge through pump is 0.04 m^3/s and the velocity of flow is constant and is equal to 2 m/s. The diameters of the suction and delivery pipes are 150 mm and 100 mm respectively. The suction and delivery heads are 6 m (abs.) and 30 m (abs.) of water respectively. If the outlet vane angle is 45° and power required to drive the pump is 16.186 kW, determine: (15)

(i) Vane angle of the impeller at inlet
(ii) The overall efficiency of the pump
(iii) Manometric efficiency of the pump

OR

VII. Explain the effect of acceleration and friction in suction and delivery pipes on indicator diagrams of a reciprocating pump. (15)

VIII. Explain with the help of neat sketch, the principle and working of the following hydraulic devices: (15)

(i) Hydraulic torque converter
(ii) Hydraulic accumulator
(iii) Hydraulic intensifier

OR

IX. (a) Explain the principle and working of hydraulic ram with a neat sketch. Also obtain the expression for Rankine's efficiency of a hydraulic ram. (8)

(b) An accumulator has a ram of diameter 250 mm and a lift of 8 m. The total weight on the accumulator is 70 kN. The packing friction is 5% of the load on the ram. Find the power delivered to the machine if the ram falls through the full height in 100 s and at the same time the pumps are delivering 0.028 m^3/s through the accumulator. (7)

***